

Autobots FOR
MAE 322
Dec. 15, 2025



Team Members:

Arthur Balva, Dwayne Bazil, Connor Brown, Jack Byrnes,
Evan Callas, Lucas Denault, Tyler Harris, Ani Kozak,
William Li, Sophia Verwoert, Sofia Zypman

(Re)Meet the Team:

Overall Team Lead: Jack Byrnes

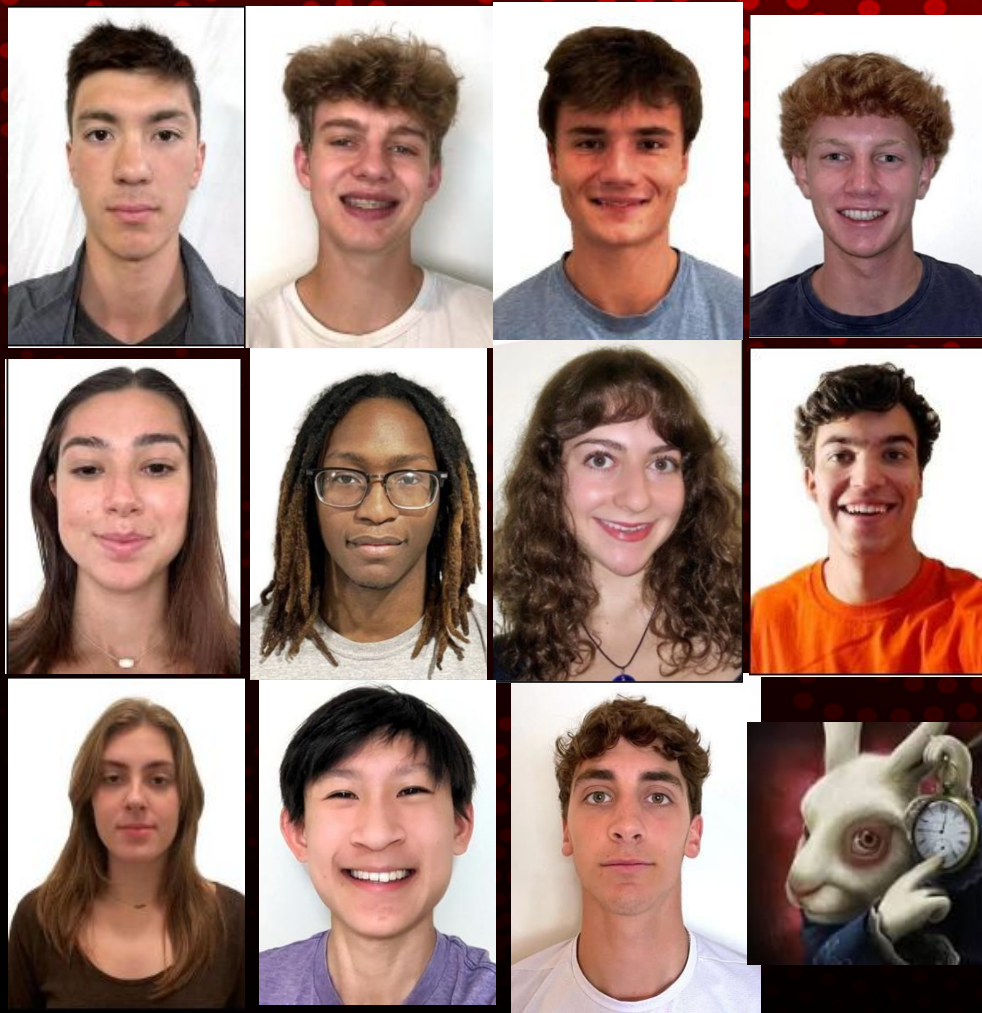
Mechanical Leads:

- CAD - Arthur Balva & Tyler Harris
- Manufacturing - Evan Callas

Electrical Lead - Lucas Denault

Software Lead - William Li

Each team member participated in various elements of robot design, manufacture, and implementation, expanding their realm of expertise.



Design Philosophy:

Lightweight - Minimize Excess Features, Save Energy

Robust Appearance - Minimal exposed components

Easy To Manufacture - Basic Parts, Minimal Machining

Low Cost - Reuse Materials In MAE Shop, and cheaper COTS Parts

Redundancy and Simplicity - Optimize Design To Have a Minimal Number Of Simple Parts.



Constraints: Cost and Weight

Costs minimized by:

- Reusing Scrap Materials Found in the Shop
- Repurpose Wheels, Electronic Components
- Borrowing Motors and Sensors from Demonstration Robot

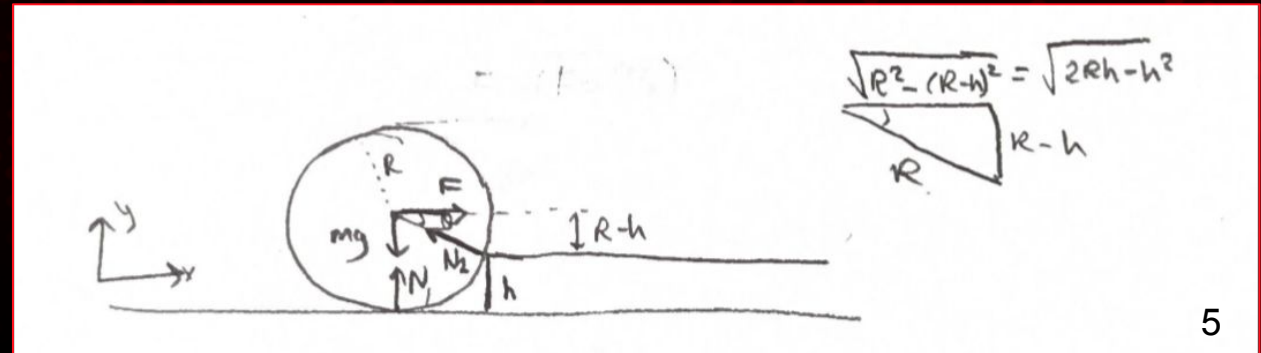
Weight Minimized by:

- Material Choices
- Motor Selection
- Conscious Design



Wall Breaching Method

- Large Wheels & Tire Attachments
- Spare tires cut up to allow tire to grab wall without slipping
- Bottom plates to increase friction to keep motor in place
- Optional Powered Back Wheel



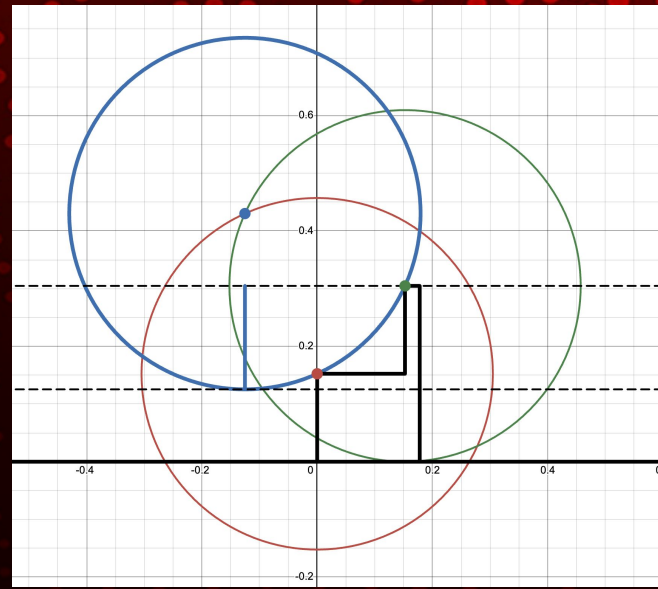
Wall Breaching Method:

Governing Equations:

$$F = \frac{mg\sqrt{2Rh_{eff}-(h_{eff})^2}}{R-h_{eff}}$$

$$T = \frac{mg\sqrt{2Rh_{eff}-(h_{eff})^2}}{R-h_{eff}} \cdot R \cdot \frac{1}{2} \cdot 1.2$$

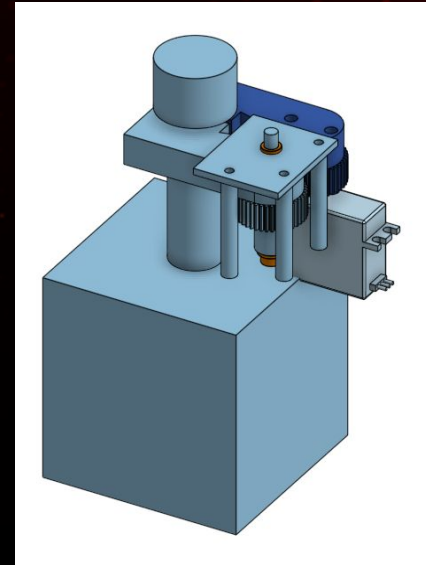
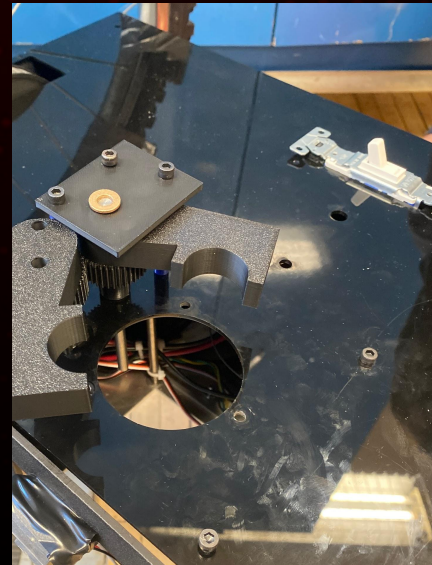
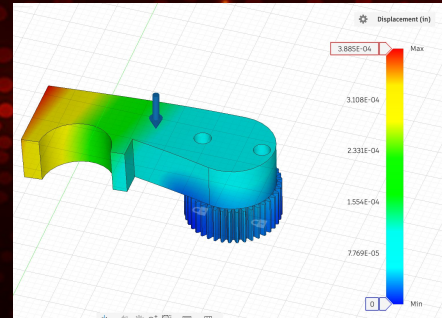
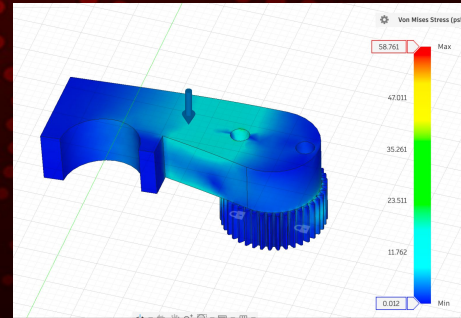
Using a factor of 1.2 as a safety margin, in order to ensure that the robot will be capable of breaching the wall.



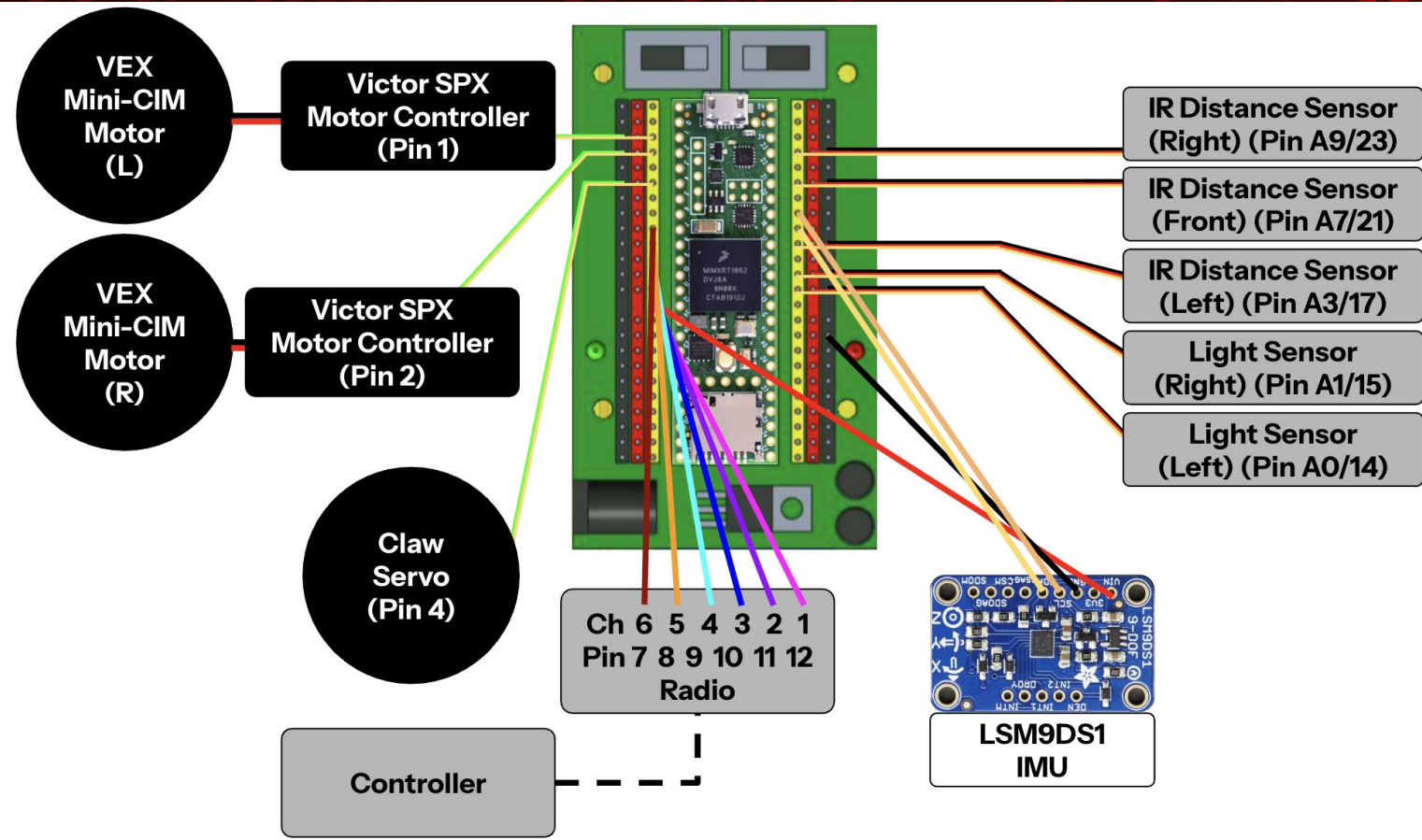
Variable	Value	Value (SI)
R	12 in	0.3048 m
h_{eff}	0.1794 m	0.1794 m
m	20 lb	9.07185 kg
T	36.01 Nm	36.01 Nm

Medkit Delivery

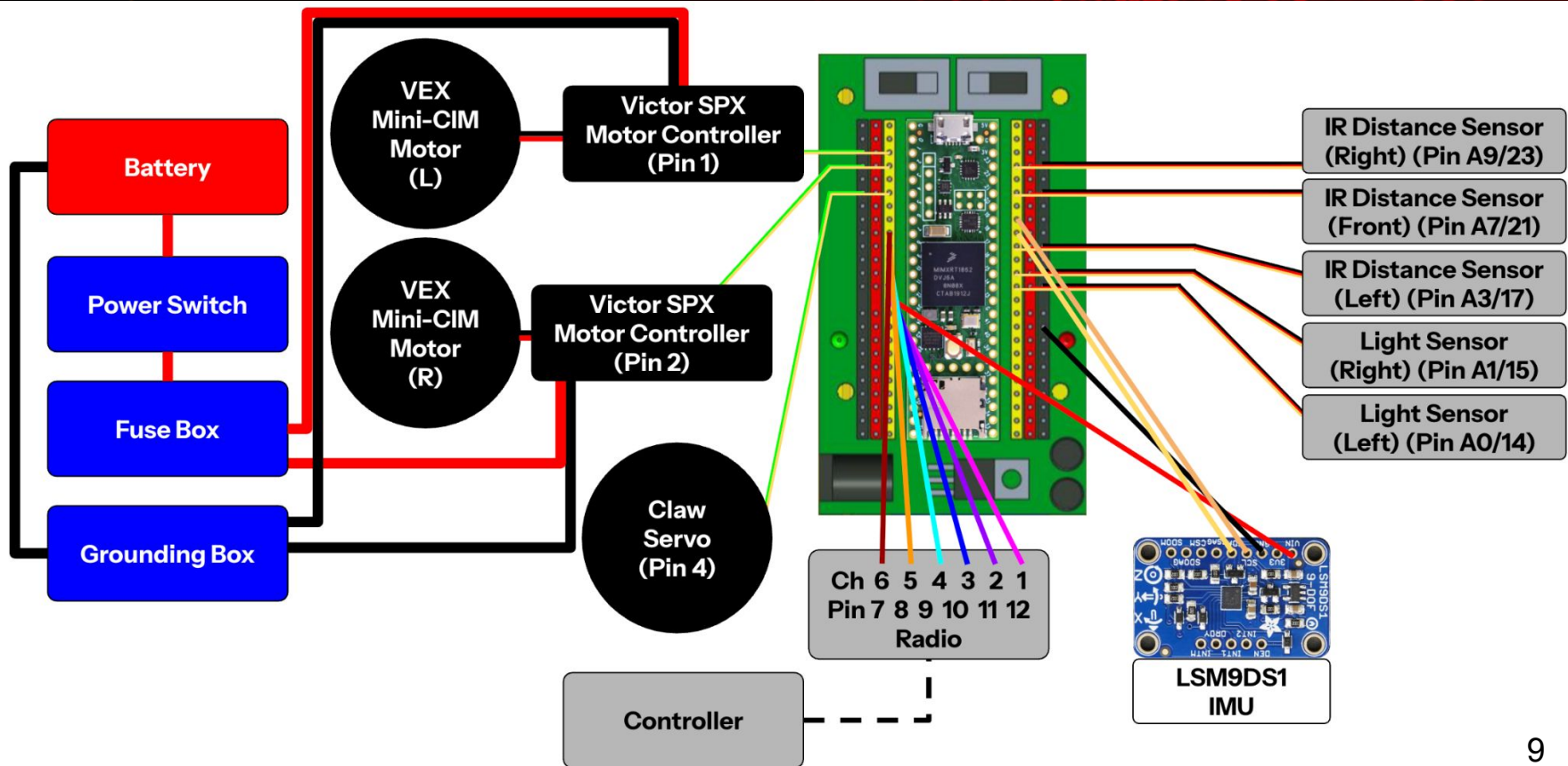
- Cylindrical Claw Holds Medkit Over Bottom Plate During transit
- Bulkhead navigates beyond light source, aligns box and center hole.
- Note Movement Forward to the Front of Bulkhead



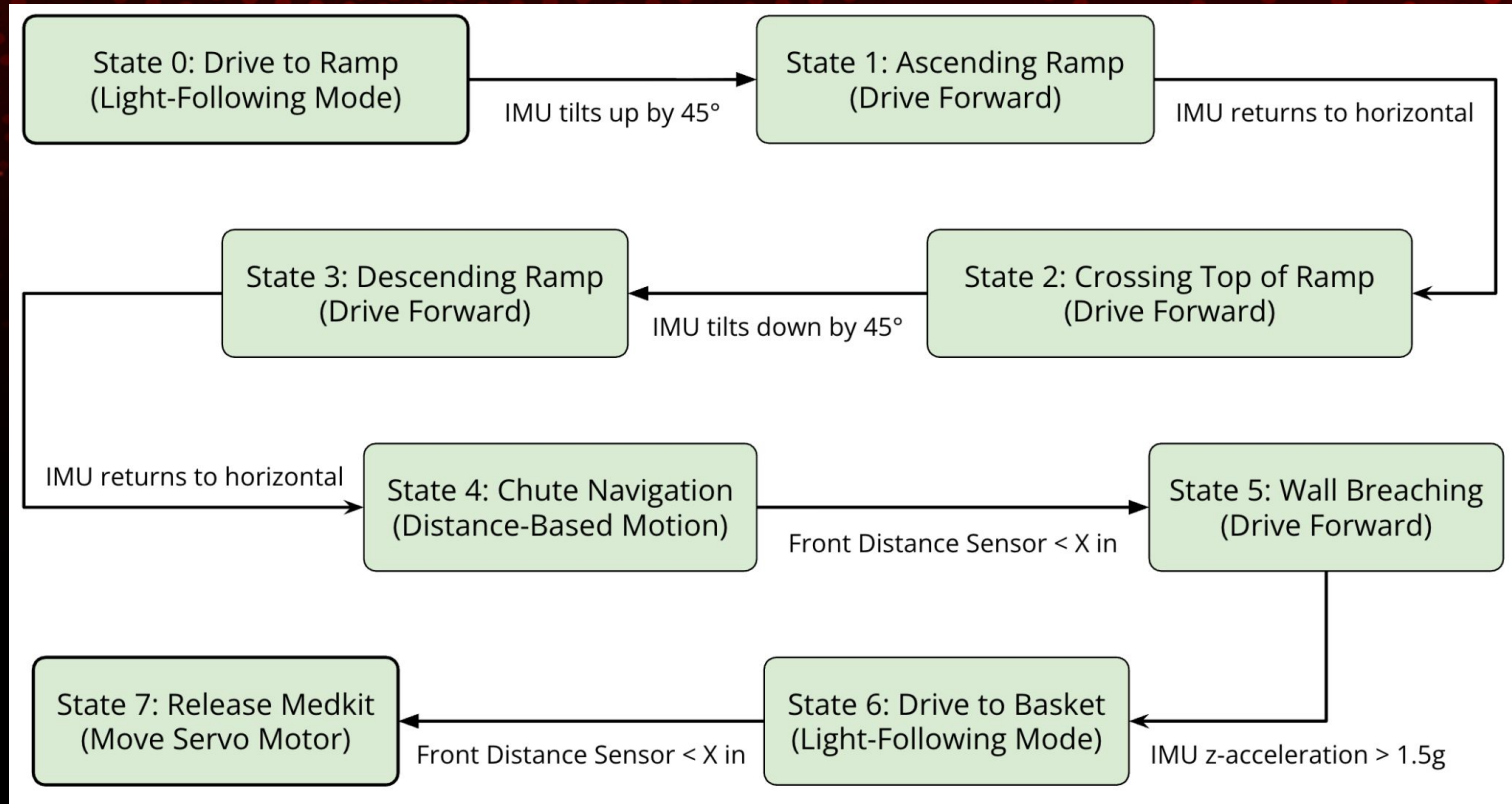
Data Passage Diagram



Full Wiring Diagram



Code Block Diagram

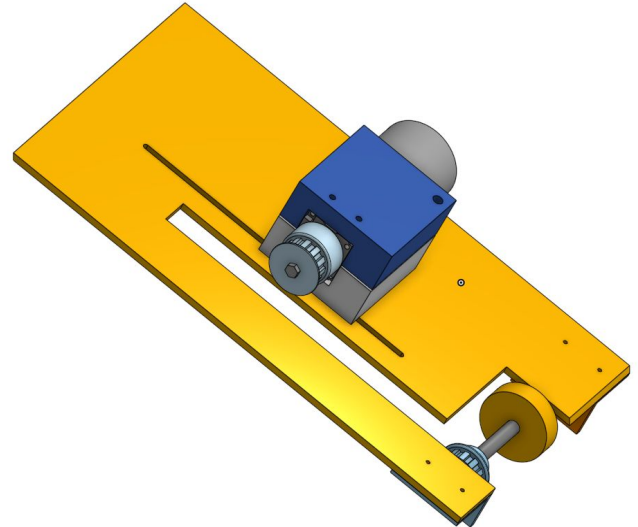
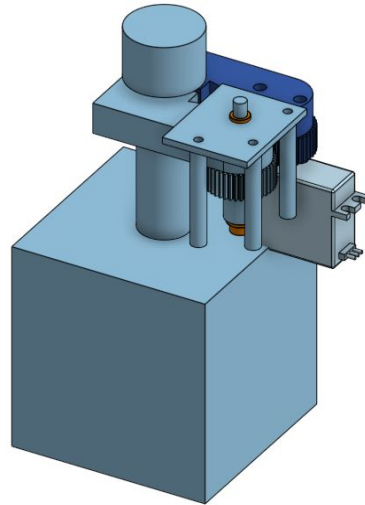
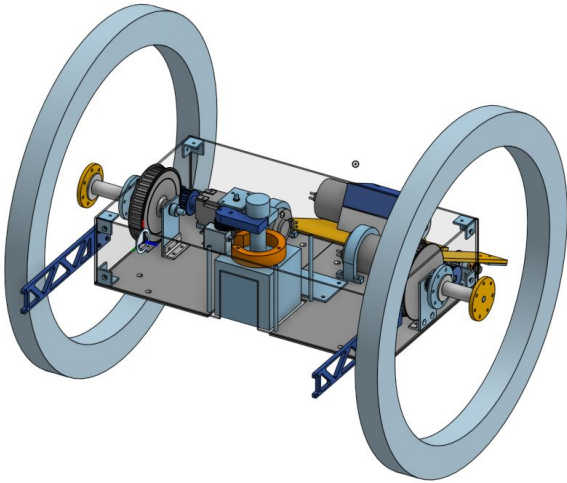
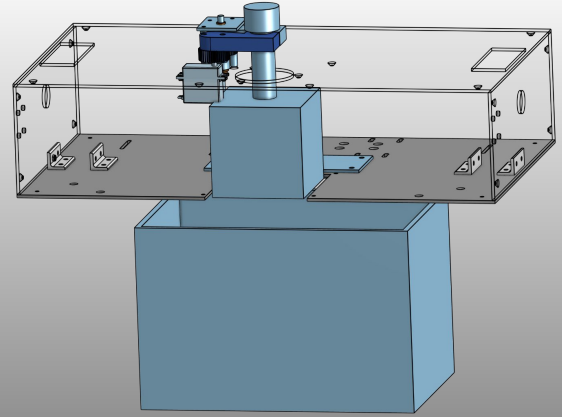


Example Code State Switching Based on Block Diagram

```
175 if(state == 0){
176     // State 0: Drive forward toward light on the ramp
177     followLight();
178     if(zoom > 2){ // Change to if gyroscope tilts up, exit followLight and state++
179         state++;
180         Serial.println("State: " + (String)state + ". Ascending ramp");
181         R_Servo.writeMicroseconds(1500); // stops motors
182         L_Servo.writeMicroseconds(1500);
183         delay(100); // Wait 100 ms before starting next state
184     }
185 }
186
187 readSensors(); // must read sensors in between state changes because the autonomous function is not looping
188
189
190 if(state == 1){
191     // State 1: Ascending the ramp
192     R_Servo.writeMicroseconds(1000);
193     L_Servo.writeMicroseconds(2000);
194     if(zoom < 2){
195         state++;
196         Serial.println("State: " + (String)state + ". Reached the top of the ramp.");
197         R_Servo.writeMicroseconds(1500); // stops motors
198         L_Servo.writeMicroseconds(1500);
199         delay(100);
200     }
201 }
202 }
```

```
203 readSensors();
204
205 if(state == 2){
206     // State 2: Driving over the top of the ramp
207     R_Servo.writeMicroseconds(1000);
208     L_Servo.writeMicroseconds(2000);
209     if(zoom > 2){
210         state++;
211         Serial.println("State: " + (String)state + ". Now descending ramp");
212         R_Servo.writeMicroseconds(1500); // stops motors
213         L_Servo.writeMicroseconds(1500);
214         delay(100);
215     }
216 }
```

CAD Drawings and Visuals:



Identified Risks:

Schedule Risks:

- Led to Crises and Shortcomings:
 - Computers Down for Two Weeks At Crucial Stage
 - Workload Led to Pushing Back Tasks

Programming Concerns:

- Chassis/Drivetrain Issues Harmed Autonomous Development:
 - Mock Chassis Insufficient Testbed

Unidentified Risks:

Medkit Drop Location:

- Required Moving Hole Forwards

Power Distribution:

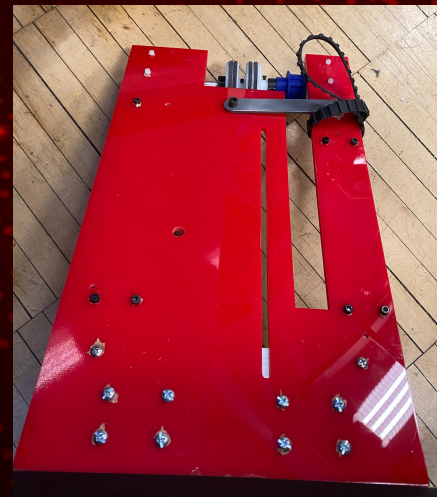
- Wheel Hub Construction:
 - Broke During Driving Operations Frequently
 - Required Iterative Solutions (Time Consuming)
- Timing Belt
 - Solvable by Introducing More Tension
 - Ran Out of Time

Bulkhead's Pros:

- Easy to manufacture/assemble and recreate
- Modular & Protected
- Built off Existing Platforms
- Adaptability (Potential Powered Rear Wheel)

Bulkhead's Cons:

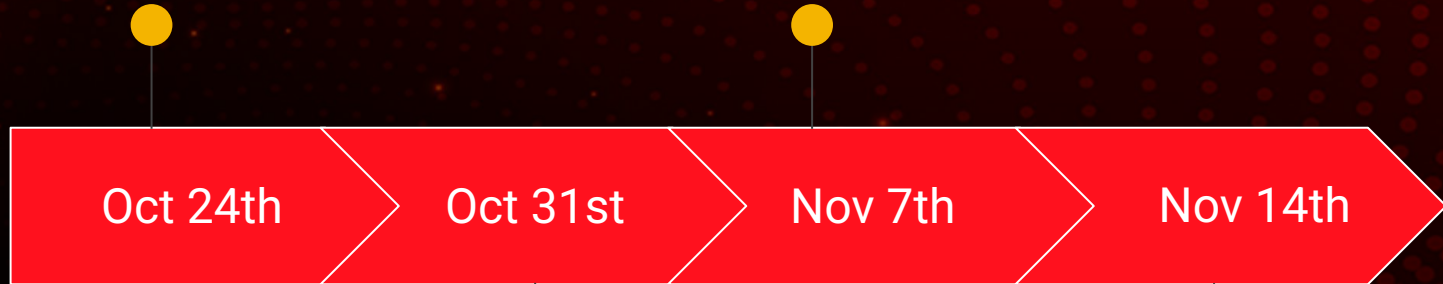
- Initial reliance on 3D Printed Materials
- Unreliable Drivetrain Tensioning
- Wide Body



Timeline/Workload Comparison

PDR completed

1. Finish Chassis
2. Wire Analysis/Assembly
3. Programming - Light
4. Programming - Wall



Key:

Completed On Time
Completed Later

Preliminary CAD
finalized

1. Bulkhead Assembled
2. Test Bulkhead
 - a. OL Breach of Wall

Timeline/Workload Comparison Continued

Thanksgiving 🦃

1. Lock it up
2. Get some rest



1. Final CAD
2. Finalize Tweaks
post test

Final Design Report

3. Successful demo of Bulkhead over entire course,
multiple times to ensure success was not a fluke

Time Summary

Likely the Harshesht Constraint:

- Late Start
- Shop Closure
- Other Coursework

Projected Man Hours: 250 hrs

Estimated Time Spent: 550 hrs

CAD: 100 hrs

Manufacturing: 190 hrs

Automation: 30 hrs

Electrical: 30 hrs

Troubleshooting/Problem Solving:
200 hrs



Summary and Conclusions:

Bulkhead Was:

- A product of collaboration
- Capable of Open Loop Operation Over the Ramp and Through Chute
- Worked on Extensively, at All Hours



Bulkhead Was Not:

- As Successful As Team Hoped
- Capable of Totally Autonomous Operation
- As Straightforward as the Team Hoped